**Leeds University School of Computing**

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# Assessed Coursework Coversheet

For use with *individual* assessed work

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| **Student ID Number:** | 2 | 0 | 2 | 0 | 0 | 5 | 6 | 6 | 0 |
| **Module Code:** | COMP5840M | | | | | | | | |
| **Module Title:** | Data Mining and Text Analytics | | | | | | | | |
| **Module Leader:** | Professor Eric Atwell | | | | | | | | |
| **Declared Word Count:** | 3183 | | | | | | | | |

**Please read the following carefully and be accurate in your responses; they are all important:**

| | **Delete as appropriate** |
|---|---|
| By submitting this work I declare it is all my own work, other than where indicated by references. I have not colluded with others, re-submitted past work of my own, submitted any work done by others or by Generative AI unless indicated, or otherwise breached the University academic integrity rules. I understand that any discrepancies between this declaration and the assignment could result in an academic malpractice procedure.<br><br>Read the full University of Leeds declaration of academic integrity here https://secretariat.leeds.ac.uk/wp-content/uploads/sites/109/2022/12/academic_integrity.pdf | YES |
| My declared word count is accurate and I have not attempted to mislead. I understand that making a fraudulent statement about word count could result in an academic malpractice procedure, and/or may impact the mark. | YES |
| I have applied for an extension but have not yet heard back on whether it is granted. I am submitting this paper in the knowledge that I may request to submit a later version when I hear back from the extension team. Markers should be aware that this may not be my final version of the assignment. | NO |
| I am aware of the GAI category for this assignment (delete opposite to show the correct one), and have adhered to the guidance for that category. | GREEN |

**Assignments should be submitted in time but will be accepted (with late penalties) up to 14 days after deadline**

Late penalties = 5 marks per 24hours late

# A Comparative Study of Traditional Machine Learning and Transformer-Based Models for Toxic Comment Detection

## 1. Research Hypothesis and Objectives

Online platforms depend heavily on user-generated content, including comments, reviews, forum posts, and social media messages. While this enables communication and participation, it also creates challenges for detecting harmful language such as abuse, harassment, insults, and hate-related content. Manual moderation is difficult to adjust, therefore automatic toxic comment detection is an important data mining and text analytics problem.

This project proposes a six-month research project to investigate whether transformer-based language models can improve toxic comment detection compared with traditional machine learning approaches. The main research hypothesis is:

Transformer-based models using contextual language representations will detect toxic comments more effectively than traditional TF-IDF-based machine learning models, particularly in terms of recall and F1-score for the toxic class.

The project is challenging because toxic language is often context-dependent. A comment may contain offensive words without being genuinely harmful, while another comment may be toxic through implication, sarcasm, or targeted abuse. This means that keyword-based or simple bag-of-words approaches may fail to capture the full meaning of a sentence. The project compares a traditional text classification pipeline with an improved class-balanced version and a transformer-based model in a small pilot study, then uses the findings to design a larger research programme.

The main objective is to design, evaluate, and compare toxic comment classification approaches using a real-world labelled dataset. The specific objectives are:

• To select and preprocess a suitable toxic comment dataset.

• To convert the original multi-label toxicity categories into a binary toxic/non-toxic classification task.

• To implement a baseline model using TF-IDF and Logistic Regression.

• To investigate the effect of class imbalance and improve the baseline using class weighting.

• To apply a transformer-based model for contextual toxic comment detection.

• To evaluate the models using precision, recall, F1-score, accuracy, and confusion matrices.

• To identify limitations, risks, and ethical issues, including false positives and potential bias in toxicity classification.

• To propose a structured six-month research programme using CRISP-DM.

The expected outcome is not only a working classifier, but also a clear understanding of the trade-offs between traditional and transformer-based methods. In particular, the project aims to determine whether higher recall for toxic comments justifies possible increases in false positives when the system is used for practical moderation.

## 2. Importance and Contribution to Knowledge

The detection of toxic comments is an increasingly essential issue because of the rapid growth in online communication platforms such as social media, forums and review systems. Harmful content, including abusive language, harassment and hate speech, can negatively affect users, reduce engagement and create unsafe online environments. As a result, automated

toxic comment detection has become a major application of data mining and natural language processing.

This research contributes to the field by exploring how effective different approaches are to text classification. Traditional machine learning methods, such as TF-IDF combined with Logistic Regression, are widely used due to their simplicity and efficiency. However, these approaches often rely on very simple representations of text and may fail to capture its meaning. In contrast, transformer-based models such as BERT use deep background representations and have demonstrated strong performance across a range of natural language processing tasks (Devlin et al., 2019).

The contribution of this project lies in providing a comparison analysis between traditional and modern approaches within the same experimental framework. By evaluating baseline, improved, and transformer-based models on the same dataset, the study highlights how different techniques perform under realistic conditions. In particular, the project focuses on identifying trade-offs between precision and recall, especially in the detection of toxic comments, where missing harmful content can have serious consequences.

Furthermore, the project contributes to practical understanding by demonstrating how class imbalance affects model performance and how simple techniques such as class weighting can improve results. The inclusion of a transformer-based model extends this analysis by showing how the main understanding can enhance detection capabilities.

Overall, this research provides both theoretical and practical contributions by linking demonstrated concepts in text classification with modern deep learning approaches, and by showing their impact through a structured pilot study.

## 3. Background

Text classification is a fundamental task in natural language processing, where the goal is to assign a piece of text to one or more predefined categories based on its content. In the context of this project, the task is to classify comments as either toxic or non-toxic. This type of problem is commonly approached using supervised machine learning, where models are trained on labelled data and then used to predict the class of new, unseen text.

Traditional approaches to text classification often rely on representing text as numerical features. One of the most widely used techniques is Term Frequency–Inverse Document Frequency (TF-IDF), which measures how important a word is within a document relative to a collection of documents. These features are then used with classification algorithms such as Logistic Regression, which is a simple yet effective method for binary classification tasks. However, these approaches mainly rely on how often some words are used and do not capture the context and where words appear.

A key challenge in classification problems, especially in tasks such as toxic comment detection, is class imbalance. In many real-world datasets, non-toxic comments significantly outnumber toxic ones. This imbalance can lead to biased models that perform well on the majority class but fail to correctly identify minority cases. Because of this, evaluation metrics such as precision, recall, and F1-score are often preferred over accuracy, as they provide a more detailed understanding of model performance.

More recently, deep learning approaches have been introduced to improve text classification performance. Transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers) are designed to understand the context of words within a sentence by considering both left and right context simultaneously. These models are pre-trained on large text corpora and can then be fine-tuned for specific tasks. “BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers” (Devlin et al., 2019).Overall, the development of text classification methods has moved from simple frequency-based representations to more

advanced contextual models. This progression is important for tasks such as toxic comment detection, where understanding the meaning and intent of a sentence is often more important than simply identifying individual words.

## 4. Critical Analysis of Existing Solutions

Toxic comment detection has been widely studied using a range of machine learning and natural language processing techniques. Traditional approaches such as Naive Bayes, Logistic Regression, and Support Vector Machines have been commonly used for text classification tasks due to their simplicity and efficiency. These methods typically rely on feature representations such as TF-IDF, which capture word frequency information but do not fully consider the context in which words appear.

One limitation of simpler models such as Naive Bayes is the assumption of feature independence. In text data, this assumption is often unrealistic, as the meaning of a sentence depends on the relationship between words rather than individual terms. As a result, such models may struggle to correctly classify more complex or context-dependent examples of toxic language.

Logistic Regression, on the other hand, is often considered a strong baseline for text classification tasks. It performs well with TF-IDF features and is less restrictive than Naive Bayes in terms of assumptions. However, like other traditional machine learning approaches, it still relies on surface-level representations of text and may fail to capture deeper semantic meaning, particularly in cases involving sarcasm, implicit toxicity, or subtle linguistic patterns.

A major challenge in toxic comment detection is class imbalance, where non-toxic comments significantly outnumber toxic ones. This imbalance can negatively affect model performance and lead to misleading evaluation results. In particular, accuracy alone may not be a reliable metric in such scenarios, as a model can achieve high accuracy simply by predicting the majority class. Therefore, evaluation metrics such as precision, recall, and F1-score are more appropriate for assessing performance in imbalanced classification tasks (Juba and Le, 2019).

Previous research also suggests that techniques designed to correct class imbalance, such as oversampling or undersampling, do not always lead to consistent improvements in performance. In many cases, increasing the size of the training data has a greater impact on improving precision and recall than applying imbalance correction methods. This highlights that the core difficulty lies not only in the imbalance itself, but also in the availability of sufficient data (Juba and Le, 2019).

More recently, deep learning approaches, particularly transformer-based models such as BERT, have been introduced to address the limitations of traditional methods. Deep learning–based models have been shown to outperform traditional machine learning approaches in a wide range of text classification tasks, due to their ability to capture more complex patterns and contextual information (Minaee et al., 2021). These models are able to capture contextual relationships between words and better understand the meaning of sentences. As a result, they are more effective in detecting subtle and context-dependent forms of toxic language. However, these models are computationally expensive and require significantly more resources and training data compared to traditional approaches.

Overall, while traditional machine learning methods provide a strong and efficient baseline, they are limited in their ability to capture context. Transformer-based models offer improved performance but introduce higher calculating cost. Therefore, selecting an appropriate method involves balancing performance, complexity, and resource requirements.

## 5. Pilot Study

This project includes a pilot study designed to evaluate the performance of different approaches for toxic comment detection. The main aim is to compare traditional machine learning methods with a transformer-based model using a real-world dataset. The study serves as an initial experiment to assess feasibility and to identify strengths and limitations of different modelling approaches before proposing a larger research plan.

Dataset used here is the Jigsaw Toxic Comment Classification dataset, which consists of user-generated comments labelled across multiple toxicity categories, including toxic, severe toxic, obscene, threat, insult, and identity hate. In order to simplify the task and make it suitable for a pilot study, the multi-label classification problem was converted into a binary classification problem. All toxicity-related labels were combined into a single “toxic” class, while comments without any of these labels were classified as “non-toxic”. This allows for a clearer comparison between different models while still preserving the key challenge of detecting harmful content.

Before training the models, the text data was preprocessed to be sure that it is consistent and improving model performance. This included converting all text to lowercase, removing punctuation, and tokenising the input text. For the traditional machine learning models, TF-IDF vectorisation was applied to transform the text into numerical features. This approach represents each document as a weighted vector based on word importance, allowing the models to process textual data effectively.

Three different models were implemented in this study. The first model is a baseline Logistic Regression classifier trained using TF-IDF features. This model serves as a simple and efficient benchmark for text classification tasks. The second model is an improved version of Logistic Regression that incorporates class weighting to address the issue of class imbalance. By assigning higher importance to the minority class, this model attempts to improve the detection of toxic comments. The third model is a transformer-based model using BERT, which captures contextual relationships between words and is expected to perform better on complex language patterns.

The models were evaluated using a range of performance metrics, including accuracy, precision, recall, and F1-score. In addition, confusion matrices were used to provide a more detailed understanding of how each model performs in distinguishing between toxic and non-toxic comments. These evaluation methods allow for a comprehensive comparison of model behaviour, particularly in the presence of class imbalance.

The results of the pilot study show clear differences in performance between the models. The baseline Logistic Regression model achieved a high overall accuracy of approximately 0.92. However, this performance is misleading when examined more closely. The recall for the toxic class was only 0.26, indicating that the model failed to correctly identify a large proportion of toxic comments. This suggests that the model is heavily biased towards predicting the majority class (non-toxic comments), which is a common issue in imbalanced datasets.

A significant improvement was observed when class weighting was applied to the Logistic Regression model. The recall for the toxic class increased from 0.26 to approximately 0.71, demonstrating a much stronger ability to detect harmful content. However, this improvement came at the cost of reduced precision, meaning that the model generated more false positives. This highlights an important trade-off between recall and precision in classification tasks, particularly in applications such as content moderation where both types of errors can have consequences.

The BERT model achieved the best overall performance among the three approaches. In particular, it achieved the highest recall of approximately 0.89 for the toxic class, along with a strong F1-score. This indicates that the model is more effective at detecting toxic comments, especially those that may rely on in text or implicit meaning. Unlike traditional models, BERT is able to capture relationships between words within a sentence, allowing it to better understand complex language patterns.

One surprising finding from the pilot study is that a model can achieve high accuracy while still performing poorly on the task. The baseline Logistic Regression model achieved an accuracy of over 90%, yet failed to detect most toxic comments. This demonstrates that accuracy alone is not a sufficient evaluation metric in imbalanced classification problems. Instead, metrics such as recall and F1-score provide a more meaningful assessment of model performance.

Another unexpected result is the extent to which class weighting improved the performance of the Logistic Regression model. While the improvement in recall was anticipated, the magnitude of the change highlights how strongly class imbalance can affect model behaviour. At the same time, the increase in false positives shows that improving one aspect of performance often leads to trade-offs in others. This reinforces the importance of selecting evaluation metrics based on the specific goals of the application.

These findings provide important insights for the future direction of the project. In applications such as toxic comment detection, recall is often more critical than precision, as failing to detect harmful content may have more serious consequences than incorrectly flagging non-toxic comments. Therefore, future work should prioritise models and techniques that maximise recall while maintaining an acceptable level of precision.

The pilot study also suggests that transformer-based models such as BERT are more suitable for this task than traditional machine learning methods. As a result, future research will focus on further improving and fine-tuning transformer-based approaches. This may include experimenting with larger datasets, adjusting model parameters, and exploring alternative architectures to enhance performance.

Overall, the pilot study demonstrates that while traditional machine learning models provide a strong baseline, they are limited in their ability to capture context. Transformer-based models offer improved performance for toxic comment detection, particularly in handling complex and context-dependent language. The finding here support the first focus of the project and provide a strong background for further research.

## 6. Project Plan

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which provides a structured and iterative approach to data mining and text analytics projects. The framework is particularly suitable for this study as it allows for systematic development, evaluation, and refinement of models for toxic comment detection.

The first phase, business understanding, involves clearly defining the problem and objectives of the project. In this case, the objective is to develop and evaluate models capable of accurately classifying comments as toxic or non-toxic. This phase also considers the real-world importance of the problem, particularly in relation to content moderation, online safety, and user experience.

The second phase, data understanding, focuses on exploring the dataset and identifying key characteristics such as class distribution, data quality, and some challenges. In this project, a major issue identified is class imbalance, where non-toxic comments are more frequent than toxic ones. Understanding this imbalance is critical for choosing the best evaluation metrics and modelling strategies.

The third phase, data preparation, involves cleaning and transforming the data to make it suitable for modelling. This includes preprocessing steps such as lowercasing, removing punctuation, and tokenisation. For traditional machine learning models, TF-IDF vectorisation is applied to convert text into numerical features, while the BERT model uses its own tokenisation and embedding mechanisms.

The fourth phase, modelling, consists of implementing and training different classification models. In this project, three models are developed: a baseline Logistic Regression model, a

class-weighted Logistic Regression model to address imbalance, and a transformer-based BERT model. These models are selected to represent both traditional and modern approaches to text classification.

The fifth phase, evaluation, assesses model performance using metrics such as accuracy, precision, recall, and F1-score. Confusion matrices are also used to analyse classification behaviour in more detail. Particular attention is given to recall for the toxic class, as missing harmful content is considered more critical than false positives in many real-world applications.

The final phase, deployment, considers how the developed models could be applied in practice. For example, the system could be integrated into social media platforms or online forums to assist in automatic moderation of user-generated content.

In terms of project planning, this research is designed as a six-month programme consisting of multiple stages. The first month focuses on background research and dataset selection. The second and third months are dedicated to data preprocessing and implementation of baseline models. The fourth month focuses on improving model performance, including handling class imbalance and experimenting with different configurations. The fifth month involves implementing and evaluating the transformer-based model. The final month is dedicated to analysis, report writing, and refinement of the research findings.

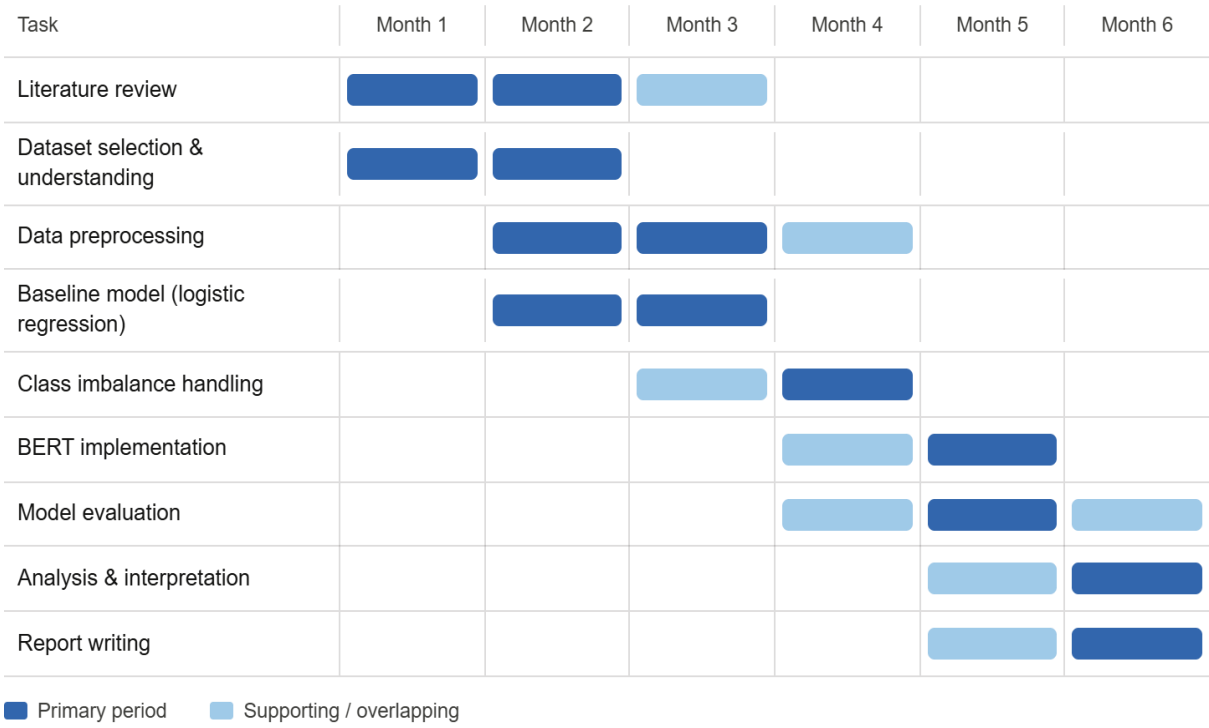
Several key checkpoints are defined throughout the project. These include successful data preprocessing, implementation of baseline models, improvement of model performance, completion of the BERT model, and final evaluation and comparison of all approaches. Each milestone represents a measurable step towards achieving the overall research objective.

The project also considers the role of stakeholders. In real-world applications, stakeholders may include platform administrators, moderators, and end users. Their needs influence the design of the system, particularly in terms of balancing detection accuracy and user fairness. For example, excessive false positives may negatively impact user experience, while low recall may allow harmful content to remain undetected.

A risk assessment is also included as part of the project plan. One key risk is class imbalance, which may lead to biased models. This is mitigated through techniques such as class weighting and the use of appropriate evaluation metrics. Another risk is overfitting, particularly in more complex models such as BERT. This can be addressed through careful validation and parameter tuning. Computational cost is also a potential challenge, especially when using transformer-based models, and may require limiting dataset size or using more efficient implementations.

Overall, the CRISP-DM framework ensures that the project follows a structured and well-defined process, from problem understanding to evaluation and potential deployment. The inclusion of clear milestones, stakeholder considerations, and risk mitigation strategies strengthens the overall research design and ensures that the project is both feasible and relevant.

## 7. Workplan Diagram



## 8. References

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Juba, B. and Le, H.S. (2019).
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Proceedings of the AAAI Conference on Artificial Intelligence.

Jigsaw Toxic Comment Classification Challenge (Kaggle, 2018)

## 9. Appendix: Use of Tools and Generative AI

### Appendix (i): Tools and Technologies

This appendix outlines the tools and technologies used throughout the development of the project.

Python was used as the primary programming language for implementing the pilot study. The pandas library was used for data loading, cleaning, and manipulation. The scikit-learn library was used to implement machine learning models such as Logistic Regression, as well as TF-IDF vectorisation and evaluation metrics.

Visualisation libraries such as matplotlib and seaborn were used to generate confusion matrices and analyse model performance. These visualisations supported the interpretation of results and helped identify classification patterns.

For the transformer-based model, the Hugging Face transformers library was used to implement a BERT-based classifier. This enabled the use of pre-trained models capable of capturing contextual relationships in text.

Jupyter Notebook was used as the development environment to organise code, run experiments, and present results clearly.

### Appendix (ii): Use of Generative AI

Generative AI tools were used to support the development of this report. ChatGPT was used to assist with structuring the report, improving clarity, and refining academic writing.

The use of AI tools was limited to support tasks such as rephrasing, summarising, and improving readability. All model implementation, data processing, and experimental analysis were carried out independently.

All AI-generated outputs were critically reviewed, edited, and adapted before being included in the final report. This ensured that the final content reflects the author's own understanding and maintains academic integrity.

### Appendix (iii): Example Prompts and Outputs

The following examples demonstrate how generative AI tools were used in three key areas of the project: the pilot study, background research, and report writing.

1. Pilot Study (Model Development)

Prompt:

"How to handle class imbalance in machine learning?"

Explanation:

The response suggested techniques such as class weighting, oversampling, and evaluation using recall and F1-score. This directly influenced the implementation of class-weighted Logistic Regression in the pilot study, which improved the model’s ability to detect toxic comments.

2. Background Research

Prompt:

"Explain the difference between precision, recall, and F1-score in classification problems."

Explanation:

The response provided clear definitions and comparisons between evaluation metrics. This supported understanding of model evaluation and helped justify the choice of metrics used in the analysis section of the report.

3. Report Writing and Presentation

Prompt:

"Improve the academic writing of this paragraph about toxic comment detection."

Explanation:

The response provided a more formal and structured version of the paragraph. This was used to improve clarity and readability in the report, while ensuring that the final content was reviewed and edited manually.

4. Additional Supporting Prompt

Prompt:

"Compare traditional machine learning models with transformer-based models in text classification."

Explanation:

The output highlighted key differences between approaches and supported the development of the critical analysis section.